

Problem 29.3: Higgs Decays

Source: M. D. Schwartz, *Quantum Field Theory and the Standard Model*, Problem 29.3.

Higgs decays.

1. Calculate the rate for $H \rightarrow b\bar{b}$ in the Standard Model.
2. Calculate the rate for $H \rightarrow gg$ in the Standard Model. The dominant contribution to this comes from a triangle loop diagram involving top quarks.
3. Calculate the rate for $H \rightarrow \gamma\gamma$ in the Standard Model. Include contributions both from top loops and from loops of W bosons.
4. Plot the branching ratios for $H \rightarrow b\bar{b}$, $H \rightarrow gg$, and $H \rightarrow \gamma\gamma$ as a function of Higgs mass.

Solution. Before symmetry breaking, the fermions couple to the Higgs field as

$$-Y_{ij}^d \bar{Q}^i H d_R^j - Y_{ij}^u \bar{Q}^i \tilde{H} u_R^j + h.c. \quad (1.1)$$

Focusing on the bottom quark, we find

$$-Y_{33}^d \begin{pmatrix} \bar{t}_L & \bar{b}_L \end{pmatrix} \begin{pmatrix} H^+ \\ H^0 \end{pmatrix} b_R + h.c. \quad (1.2)$$

Expanding the Higgs around the vacuum as usual, we find the bottom quark couples to the Higgs field $h(x)$ through

$$-\frac{y}{\sqrt{2}} \bar{b}_L h b_R - \frac{y^*}{\sqrt{2}} \bar{b}_R h b_L = -\frac{1}{\sqrt{2}} (\text{Re } y) h \bar{b} b - \frac{i}{\sqrt{2}} (\text{Im } y) h \bar{b} \gamma^5 b, \quad (1.3)$$

where we defined $y \equiv Y_{33}^d$ and we have assumed y to be complex for generality. Therefore, at tree-level the scattering amplitude is given by

$$i\mathcal{M}_b = -\frac{1}{\sqrt{2}} i (\text{Re } y) [\bar{u}(p_2) v(p_3)] + \frac{1}{\sqrt{2}} (\text{Im } y) [\bar{u}(p_2) \gamma^5 v(p_3)]. \quad (1.4)$$

Then, we find

$$\begin{aligned} \sum_{\text{spin/colour}} |\mathcal{M}_b|^2 &= \frac{3}{2} \sum_{\text{spin}} \left[-(\text{Re } y) [\bar{u}(p_2) v(p_3)] - i (\text{Im } y) [\bar{u}(p_2) \gamma^5 v(p_3)] \right] \\ &\quad \times \left[-(\text{Re } y) [\bar{v}(p_3) u(p_2)] - i (\text{Im } y) [\bar{v}(p_3) \gamma^5 u(p_2)] \right] \\ &= \frac{3}{2} [(\text{Re } y)^2 + (\text{Im } y)^2] \text{Tr}((\not{p}_2 + m_b)(\not{p}_3 - m_b)) \\ &= 6|y|^2 (p_2 \cdot p_3 - m_b^2). \end{aligned} \quad (1.5)$$

In the mass basis, y can be chosen real. To compute the decay rate in the centre-of-mass frame, we use

$$\Gamma = \frac{1}{32\pi^2 m_h^2} |\mathbf{p}| \int d\Omega |\mathcal{M}_b|^2. \quad (1.6)$$

Using

$$s = m_h^2, \quad |\mathbf{p}| = \frac{1}{2} \sqrt{m_h^2 - 4m_b^2}, \quad p_2 \cdot p_3 = \frac{m_h^2 - 2m_b^2}{2}, \quad (1.7)$$

we find

$$\Gamma(H \rightarrow b\bar{b}) = \frac{3y^2}{16\pi m_h^2} (m_h^2 - 4m_b^2)^{3/2} = \frac{3m_b^2 m_h}{8\pi v^2} \left(1 - \frac{4m_b^2}{m_h^2}\right)^{3/2}. \quad (1.8)$$

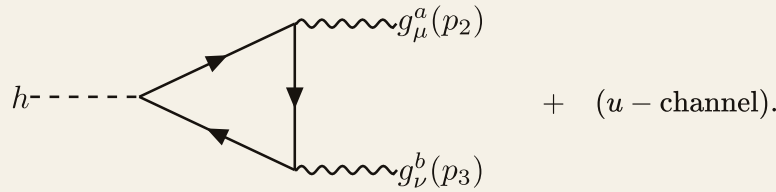
Using

$$m_h = 125.20 \text{ GeV}, \quad m_b = 4.183 \text{ GeV}, \quad v = 246.22 \text{ GeV}, \quad (1.9)$$

we obtain

$$\Gamma(H \rightarrow b\bar{b}) \simeq 4.28 \text{ MeV}. \quad (1.10)$$

There is no tree-level contribution to $H \rightarrow gg$. The lowest-order contribution comes from the quark triangle diagram



$$+ (u - \text{channel}). \quad (1.11)$$

For a particular quark flavour q , the scattering amplitude is

$$\begin{aligned} i\mathcal{M}_q^{ab} &= - \left(-i \frac{y_q}{\sqrt{2}} \right) (ig_s)^2 \int \frac{d^d k}{(2\pi)^d} \\ &\times \text{Tr} \left[\frac{i(\not{k} + m_q)}{k^2 - m_q^2} \gamma^\mu T^a \frac{i(\not{k} - \not{p}_2 + m_q)}{(k - p_2)^2 - m_q^2} \gamma^\nu T^b \frac{i(\not{k} - \not{p}_1 + m_q)}{(k - p_1)^2 - m_q^2} \right] \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \\ &+ (p_2, \mu, a \leftrightarrow p_3, \nu, b) \\ &= i \frac{y_q g_s^2}{\sqrt{2}} T_F \delta^{ab} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) 8m_q [F_q^{\mu\nu}(p_1, p_2) + F_q^{\nu\mu}(p_1, p_3)], \end{aligned} \quad (1.12)$$

where $y_q = \frac{\sqrt{2}m_q}{v} = 2\sqrt{\lambda} \frac{m_q}{m_h}$, and $F_q^{\mu\nu}$ is defined to include the factor of i below, so that it is real for the top-quark loop. Since $y_q \propto m_q$ and $m_b, m_c, m_s, m_d, m_u \ll m_h$, the dominant contribution comes from the top-quark loop. The complete amplitude is obtained by summing over quark flavours. Now, we evaluate the loop integral. The trace can be simplified as

$$\begin{aligned}
& \text{Tr} \left[\frac{i(\not{k} + m_q)}{k^2 - m_q^2} \gamma^\mu \frac{i(\not{k} - \not{p}_2 + m_q)}{(k - p_2)^2 - m_q^2} \gamma^\nu \frac{i(\not{k} - \not{p}_1 + m_q)}{(k - p_1)^2 - m_q^2} \right] \\
&= -\frac{4im_q}{(k^2 - m_q^2)[(k - p_2)^2 - m_q^2][(k - p_1)^2 - m_q^2]} \left[k^\mu(k - p_2)^\nu + k^\nu(k - p_2)^\mu - k \cdot (k - p_2) \eta^{\mu\nu} \right. \\
&\quad + k^\mu(k - p_1)^\nu + k^\nu(k - p_1)^\mu + k \cdot (k - p_1) \eta^{\mu\nu} + (k - p_2)^\mu(k - p_1)^\nu + (k - p_1)^\mu(k - p_2)^\nu \\
&\quad \left. - (k - p_2) \cdot (k - p_1) \eta^{\mu\nu} + m_q^2 \eta^{\mu\nu} \right]. \tag{1.13}
\end{aligned}$$

Let

$$A = k^2 - m_q^2, \quad B = (k - p_2)^2 - m_q^2, \quad C = (k - p_1)^2 - m_q^2. \tag{1.14}$$

Introducing Feynman parameter,

$$\frac{1}{ABC} = 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{[xA + yB + zC]^3}, \quad z = 1 - x - y, \tag{1.15}$$

the denominator becomes

$$\begin{aligned}
xA + yB + zC &= k^2 - 2k \cdot (yp_2 + zp_1) + yp_2^2 + zp_1^2 - m_q^2 \\
&= (k - yp_2 - zp_1)^2 - \Delta,
\end{aligned} \tag{1.16}$$

where

$$\Delta = m_q^2 - y(1 - y)p_2^2 - z(1 - z)p_1^2 + 2yz p_1 \cdot p_2. \tag{1.17}$$

Shifting the loop momentum, $\ell = k - yp_2 - zp_1 \equiv k - P$, the numerator becomes

$$\begin{aligned}
\mathcal{N}^{\mu\nu} &= 4\ell^\mu \ell^\nu - \ell^2 \eta^{\mu\nu} + 4P^\mu P^\nu - 2P^\mu(p_1 + p_2)^\nu - 2P^\nu p_2^\mu \\
&\quad + p_1^\mu p_2^\nu + p_2^\mu p_1^\nu + (m_q^2 - P^2 + 2P \cdot p_2 - p_1 \cdot p_2) \eta^{\mu\nu} \\
&\quad + 4\ell^\mu P^\nu + 4P^\mu \ell^\nu - 2\ell^\mu(p_1 + p_2)^\nu - 2\ell^\nu p_2^\mu + 2\ell \cdot (p_2 - P) \eta^{\mu\nu}.
\end{aligned} \tag{1.18}$$

The loop integral becomes

$$F_q^{\mu\nu}(p_1, p_2) = i \int_0^1 dx \int_0^{1-x} dy \int \frac{d^d \ell}{(2\pi)^d} \frac{\mathcal{N}^{\mu\nu}(\ell, P)}{(\ell^2 - \Delta)^3}. \tag{1.19}$$

The terms with an odd number of ℓ in the numerator vanish, and $\ell^\mu \ell^\nu \rightarrow \frac{1}{d} \eta^{\mu\nu} \ell^2$, which gives

$$\begin{aligned}
\mathcal{N}^{\mu\nu} &\longrightarrow \left(\frac{4}{d} - 1 \right) \ell^2 \eta^{\mu\nu} + 4P^\mu P^\nu - 2P^\mu(p_1 + p_2)^\nu - 2P^\nu p_2^\mu \\
&\quad + p_1^\mu p_2^\nu + p_2^\mu p_1^\nu + (m_q^2 - P^2 + 2P \cdot p_2 - p_1 \cdot p_2) \eta^{\mu\nu}.
\end{aligned} \tag{1.20}$$

Then, using dimensional regularisation expanding around $d = 4 - \epsilon$, we find

$$\begin{aligned}
F_q^{\mu\nu}(p_1, p_2) &= \frac{-1}{32\pi^2} \int_0^1 dx \int_0^{1-x} dy \frac{1}{m_q^2 - xzm_h^2} \left[m_h^2 \left(\frac{1}{2} - 2xz \right) \eta^{\mu\nu} \right. \\
&\quad - 2z(2z - 1)p_1^\mu p_1^\nu + 4y(1 - y)p_2^\mu p_2^\nu \\
&\quad - (4yz - 2z + 1)p_1^\mu p_2^\nu \\
&\quad \left. - (4yz - 2y - 2z + 1)p_2^\mu p_1^\nu \right].
\end{aligned} \tag{1.21}$$

Contracting with the on-shell gluon polarisations, it follows that

$$\begin{aligned}\epsilon_\mu^*(p_2)\epsilon_\nu^*(p_3)p_1^\mu p_1^\nu &= \epsilon_\mu^*(p_2)\epsilon_\nu^*(p_3)p_1^\mu p_2^\nu \\ &= [\epsilon^*(p_2) \cdot p_3] [\epsilon^*(p_3) \cdot p_2],\end{aligned}\tag{1.22}$$

while all terms proportional to p_2^μ vanish. Therefore,

$$\begin{aligned}\epsilon_\mu^*(p_2)\epsilon_\nu^*(p_3)F_t^{\mu\nu}(p_1, p_2) &= -\frac{1}{32\pi^2} \int_0^1 dx \int_0^{1-x} dy \frac{1-4xz}{m_t^2 - xzm_h^2} \\ &\times \left[\frac{m_h^2}{2} \epsilon^*(p_2) \cdot \epsilon^*(p_3) - (\epsilon^*(p_2) \cdot p_3) (\epsilon^*(p_3) \cdot p_2) \right].\end{aligned}\tag{1.23}$$

The crossed orientation gives the same result after relabelling the Feynman parameters. Adding the two orientations gives

$$\begin{aligned}\epsilon_\mu^*(p_2)\epsilon_\nu^*(p_3) [F_t^{\mu\nu}(p_1, p_2) + F_t^{\nu\mu}(p_1, p_3)] &= -\frac{1}{16\pi^2} \left[\frac{m_h^2}{2} \epsilon^*(p_2) \cdot \epsilon^*(p_3) - (\epsilon^*(p_2) \cdot p_3) (\epsilon^*(p_3) \cdot p_2) \right] \\ &\times \int_0^1 dx \int_0^{1-x} dy \frac{1-4x(1-x-y)}{m_t^2 - m_h^2 x(1-x-y)}.\end{aligned}\tag{1.24}$$

Evaluating the integral gives

$$\begin{aligned}\int_0^1 dx \int_0^{1-x} dy \frac{1-4x(1-x-y)}{m_t^2 - m_h^2 x(1-x-y)} &= \frac{2}{m_h^2} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right].\end{aligned}\tag{1.25}$$

Thus, the top-loop amplitude becomes

$$\begin{aligned}i\mathcal{M}_t^{ab} &= -i \frac{y_t g_s^2 m_t T_F}{2\sqrt{2}\pi^2} \delta^{ab} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right] \\ &\times \epsilon_\mu^*(p_2)\epsilon_\nu^*(p_3) \left[\eta^{\mu\nu} - \frac{2}{m_h^2} p_3^\mu p_2^\nu \right].\end{aligned}\tag{1.26}$$

Using

$$\sum_{\text{pols.}} \epsilon_\mu^*(p)\epsilon_\nu(p) = -\eta_{\mu\nu}, \quad \sum_{a,b} \delta^{ab}\delta^{ab} = N_c^2 - 1 = 8, \quad T_F = \frac{1}{2},\tag{1.27}$$

and

$$\left[\eta^{\mu\nu} - \frac{2}{m_h^2} p_3^\mu p_2^\nu \right] \left[\eta_{\mu\nu} - \frac{2}{m_h^2} p_{3\mu} p_{2\nu} \right] = 2,\tag{1.28}$$

we finally obtain

$$\sum_{\text{pols./colours}} |\mathcal{M}_t|^2 = \frac{m_t^2 y_t^2 g_s^4}{2\pi^4} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right]^2. \quad (1.29)$$

Here we have neglected the contributions from the remaining quarks.

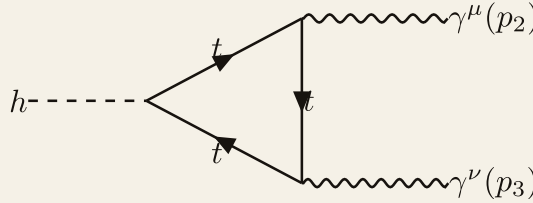
Plugging into (1.6), we find

$$\Gamma(H \rightarrow gg) = \frac{\lambda g_s^4 m_t^4}{16\pi^5 m_h^3} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right]^2, \quad (1.30)$$

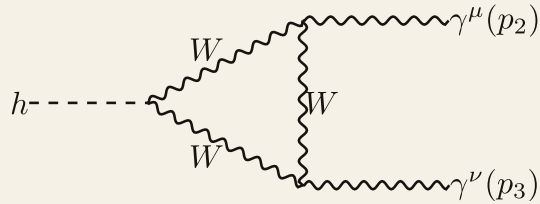
where a factor of 1/2 is added since the outgoing gluon states are identical. This gives a numerical value of

$$\Gamma(H \rightarrow gg) \simeq 1.97 \times 10^{-4} \text{ GeV} = 0.197 \text{ MeV}. \quad (1.31)$$

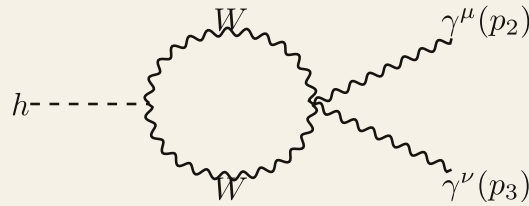
Now we turn to the computation of $H \rightarrow \gamma\gamma$. Similarly to the gluon case, there is no tree-level contribution to the process. The dominant quark-loop contribution comes from the top-quark loop, just as for the gluon case. On top of that, there is also a contribution from a W -boson loop. Therefore, our job is to compute the following diagrams



$$+ (p_2, \mu \leftrightarrow p_3, \nu) = i\mathcal{M}_t^\gamma. \quad (1.32)$$



$$+ (p_2, \mu \leftrightarrow p_3, \nu) = i\mathcal{M}_W, \quad (1.33)$$

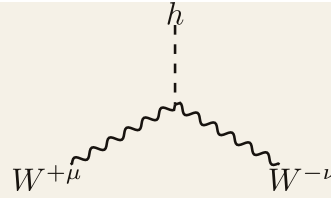


$$= i\mathcal{M}_{W,4}^{\mu\nu}. \quad (1.34)$$

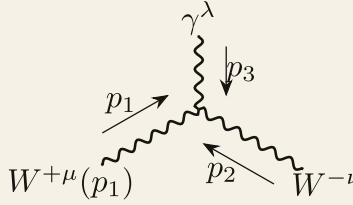
The relevant interactions are

$$\begin{aligned} \mathcal{L}_{\text{int}} = & -ie[F^{\mu\nu}W_\mu^+W_\nu^- - (\partial_\mu W_\nu^+ - \partial_\nu W_\mu^+)A^\mu(W^-)^\nu + (\partial_\mu W_\nu^- - \partial_\nu W_\mu^-)A^\mu(W^+)^\nu] \\ & + 2\frac{h}{v}m_W^2W_\mu^+W_\mu^- + e^2(A^\mu W_\mu^+A^\nu W_\nu^- - A_\mu^2W_\nu^+(W^-)^\nu) - \frac{y_t}{\sqrt{2}}h\bar{t}t + \frac{2e}{3}A_\mu\bar{t}\gamma^\mu t, \end{aligned} \quad (1.35)$$

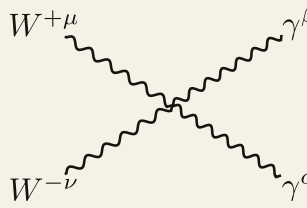
where the first three interactions are given by the diagrams



$$= i \frac{e m_W}{\sin \theta_w} \eta^{\mu\nu}, \quad (1.36)$$



$$= -ie[t] \left[\eta^{\mu\nu} (p_1 - p_2)^\lambda + \eta^{\nu\lambda} (p_2 - p_3)^\mu + \eta^{\lambda\mu} (p_3 - p_1)^\nu \right], \quad (1.37)$$



$$= ie^2[t] \left[\eta^{\alpha\mu} \eta^{\beta\nu} + \eta^{\alpha\nu} \eta^{\beta\mu} - 2\eta^{\alpha\beta} \eta^{\mu\nu} \right]. \quad (1.38)$$

and the last two are Yukawa and QED interactions. Note that $i\mathcal{M}_t^\gamma$ is almost identical to $i\mathcal{M}_t$, but now the colour trace gives $N_c = 3$ and $g_s \rightarrow 2e/3$. Therefore, we can read off from (1.26),

$$i\mathcal{M}_t^\gamma = -i \frac{2y_t e^2 m_t}{3\sqrt{2}\pi^2} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right] \times \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \left[\eta^{\mu\nu} - \frac{2}{m_h^2} p_3^\mu p_2^\nu \right]. \quad (1.39)$$

Although rather tedious, we can also read off the W bosons loop diagram from the Feynman rules (or one can easily obtain the expression using the power of AI). In unitary gauge, the W -boson propagator is

$$D_W^{\alpha\beta}(q) = \frac{-iP^{\alpha\beta}(q)}{q^2 - m_W^2}, \quad P^{\alpha\beta}(q) \equiv \eta^{\alpha\beta} - \frac{q^\alpha q^\beta}{m_W^2}. \quad (1.40)$$

It is useful to define the momentum-dependent part of the $\gamma W^+ W^-$ vertex by

$$\Gamma^{\lambda\alpha\beta}(q, r, s) = \eta^{\alpha\beta}(r - s)^\lambda + \eta^{\beta\lambda}(s - q)^\alpha + \eta^{\lambda\alpha}(q - r)^\beta, \quad (1.41)$$

so that the vertex is $-ie\Gamma^{\lambda\alpha\beta}(q, r, s)$.

The triangle diagram and its crossed photon ordering together give

$$i\mathcal{M}_W = \frac{e^3 m_W}{\sin \theta_w} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \int \frac{d^d k}{(2\pi)^d} \left[\frac{\mathcal{N}^{\mu\nu}(k; p_2, p_3)}{(k^2 - m_W^2)[(k - p_2)^2 - m_W^2][(k - p_1)^2 - m_W^2]} + \frac{\mathcal{N}^{\nu\mu}(k; p_3, p_2)}{(k^2 - m_W^2)[(k - p_3)^2 - m_W^2][(k - p_1)^2 - m_W^2]} \right], \quad (1.42)$$

where

$$\begin{aligned}\mathcal{N}^{\mu\nu}(k; p_2, p_3) &= \eta^{\alpha\rho} P_{\alpha\beta}(k) \Gamma^{\mu\beta\gamma}(-p_2, k, p_2 - k) \\ &\times P_{\gamma\delta}(k - p_2) \Gamma^{\nu\delta\sigma}(-p_3, k - p_2, p_1 - k) P_{\sigma\rho}(k - p_1).\end{aligned}\quad (1.43)$$

The quartic $\gamma\gamma W^+ W^-$ bubble diagram must also be included. Define

$$Q^{\mu\nu\beta\sigma} = \eta^{\mu\beta} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\beta} - 2\eta^{\mu\nu} \eta^{\beta\sigma}, \quad (1.44)$$

so that the quartic vertex is $ie^2 Q^{\mu\nu\beta\sigma}$. Its contribution is

$$i\mathcal{M}_{W,4} = \frac{e^3 m_W}{\sin\theta_w} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \int \frac{d^d k}{(2\pi)^d} \frac{\eta^{\alpha\rho} P_{\alpha\beta}(k) Q^{\mu\nu\beta\sigma} P_{\sigma\rho}(k - p_1)}{(k^2 - m_W^2)[(k - p_1)^2 - m_W^2]}. \quad (1.45)$$

Now, we shall evaluate the two diagrams. Introduce Feynman parameters,

$$\begin{aligned}\frac{1}{D_0 D_2 D_1} &= 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{[xD_0 + yD_2 + zD_1]^3}, \\ \frac{1}{D_0 D_3 D_1} &= 2 \int_0^1 dx \int_0^{1-x} dy \frac{1}{[xD_0 + yD_3 + zD_1]^3}, \quad z = 1 - x - y,\end{aligned}\quad (1.46)$$

where

$$\begin{aligned}D_0 &= k^2 - m_W^2, & D_1 &= (k - p_1)^2 - m_W^2, \\ D_2 &= (k - p_2)^2 - m_W^2, & D_3 &= (k - p_3)^2 - m_W^2.\end{aligned}\quad (1.47)$$

For the first and crossed orientations, respectively, make the shifts

$$\ell = k - yp_2 - zp_1, \quad \ell = k - yp_3 - zp_1. \quad (1.48)$$

Using

$$p_1 = p_2 + p_3, \quad p_1^2 = m_h^2, \quad p_2^2 = p_3^2 = 0, \quad (1.49)$$

both denominators reduce to

$$xD_0 + yD_{2,3} + zD_1 = \ell^2 - \Delta, \quad \Delta = m_W^2 - xzm_h^2. \quad (1.50)$$

Therefore, the two triangle orientations together become

$$\begin{aligned}i\mathcal{M}_W &= 2 \frac{e^3 m_W}{\sin\theta_w} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \int_0^1 dx \int_0^{1-x} dy \int \frac{d^d \ell}{(2\pi)^d} \\ &\times \frac{\mathcal{N}^{\mu\nu}(\ell + yp_2 + zp_1; p_2, p_3) + \mathcal{N}^{\nu\mu}(\ell + yp_3 + zp_1; p_3, p_2)}{(\ell^2 - \Delta)^3}.\end{aligned}\quad (1.51)$$

For convenience, define the combined numerator

$$\begin{aligned}\widetilde{\mathcal{N}}^{\mu\nu}(\ell; x, y) &\equiv \mathcal{N}^{\mu\nu}(\ell + yp_2 + zp_1; p_2, p_3) \\ &+ \mathcal{N}^{\nu\mu}(\ell + yp_3 + zp_1; p_3, p_2).\end{aligned}\quad (1.52)$$

Thus,

$$i\mathcal{M}_W = 2 \frac{e^3 m_W}{\sin \theta_w} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \int_0^1 dx \int_0^{1-x} dy \int \frac{d^d \ell}{(2\pi)^d} \frac{\widetilde{\mathcal{N}}^{\mu\nu}(\ell; x, y)}{(\ell^2 - \Delta)^3}. \quad (1.53)$$

To ensure convergence we should include the four-point diagram before carrying out the loop integral. The terms with odd l in the numerator vanishes and $l^\mu l^\nu \rightarrow \frac{1}{d} \eta^{\mu\nu} l^2$, the l^4 term in $i\mathcal{M}_{W,4}$ becomes

$$\ell^\mu \ell^\nu \ell^\rho \ell^\sigma \longrightarrow \frac{\ell^4}{d(d+2)} \left(\eta^{\mu\nu} \eta^{\rho\sigma} + \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho} \right). \quad (1.54)$$

After some long and cumbersome algebra, one would find

$$i\mathcal{M}_W + i\mathcal{M}_{W,4} = i \frac{e^3}{16\pi^2 m_W \sin \theta_w} \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \left[(p_2 \cdot p_3) \eta^{\mu\nu} - p_3^\mu p_2^\nu \right] \times \left[2 + 12m_W^2 \int_0^1 dx \int_0^{1-x} dy \frac{1-2xz}{m_W^2 - m_h^2 xz} \right]. \quad (1.55)$$

The integral can be evaluated, which gives

$$\int_0^1 dx \int_0^{1-x} dy \frac{1-2xz}{m_W^2 - m_h^2 xz} = \frac{1}{m_h^2} \left[1 + \left(2 - \frac{4m_W^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_W} \right) \right]. \quad (1.56)$$

Finally, we end up with

$$i\mathcal{M}_W + i\mathcal{M}_{W,4} = i \frac{e^3}{16\pi^2 m_W \sin \theta_w} \left[2 + \frac{12m_W^2}{m_h^2} + \frac{24m_W^2}{m_h^2} \left(1 - \frac{2m_W^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_W} \right) \right] \times \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \left[(p_2 \cdot p_3) \eta^{\mu\nu} - p_3^\mu p_2^\nu \right]. \quad (1.57)$$

To combine the top and W contributions, define

$$\begin{aligned} \mathcal{F}_t &= \frac{32m_t^2}{3m_h^2} \left[1 + \left(1 - \frac{4m_t^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_t} \right) \right], \\ \mathcal{F}_W &= 2 + \frac{12m_W^2}{m_h^2} + \frac{24m_W^2}{m_h^2} \left(1 - \frac{2m_W^2}{m_h^2} \right) \arcsin^2 \left(\frac{m_h}{2m_W} \right). \end{aligned} \quad (1.58)$$

Using $y_t = \sqrt{2}m_t/v$ and $m_W = ev/(2 \sin \theta_w)$, the total amplitude is

$$i\mathcal{M}_{\gamma\gamma} = i \frac{e^3}{16\pi^2 m_W \sin \theta_w} (\mathcal{F}_W - \mathcal{F}_t) \epsilon_\mu^*(p_2) \epsilon_\nu^*(p_3) \left[(p_2 \cdot p_3) \eta^{\mu\nu} - p_3^\mu p_2^\nu \right]. \quad (1.59)$$

Thus, using $p_2 \cdot p_3 = m_h^2/2$, the polarization sum gives

$$\begin{aligned} \sum_{\text{pols.}} |\mathcal{M}_{\gamma\gamma}|^2 &= \frac{e^6}{256\pi^4 m_W^2 \sin^2 \theta_w} (\mathcal{F}_W - \mathcal{F}_t)^2 \left[(p_2 \cdot p_3) \eta^{\mu\nu} - p_3^\mu p_2^\nu \right] \left[(p_2 \cdot p_3) \eta_{\mu\nu} - p_{3\mu} p_{2\nu} \right] \\ &= \frac{e^6 m_h^4}{512\pi^4 m_W^2 \sin^2 \theta_w} (\mathcal{F}_W - \mathcal{F}_t)^2. \end{aligned} \quad (1.60)$$

The two photons are identical, so the two-body phase space includes a factor of $1/2!$. Therefore,

$$\Gamma(H \rightarrow \gamma\gamma) = \frac{e^6 m_h^3}{16384 \pi^5 m_W^2 \sin^2 \theta_w} (\mathcal{F}_W - \mathcal{F}_t)^2 = \frac{\alpha^2 m_h^3}{256 \pi^3 v^2} (\mathcal{F}_W - \mathcal{F}_t)^2. \quad (1.61)$$

Using

$$\begin{aligned} m_h &= 125.20 \text{ GeV}, & m_t &= 172.56 \text{ GeV}, & m_W &= 80.377 \text{ GeV}, \\ v &= 246.22 \text{ GeV}, & \alpha^{-1} &= 137.036, \end{aligned} \quad (1.62)$$

we find

$$\mathcal{F}_t \simeq 1.836, \quad \mathcal{F}_W \simeq 8.331, \quad (1.63)$$

and hence

$$\Gamma(H \rightarrow \gamma\gamma) \simeq 9.16 \times 10^{-6} \text{ GeV} = 9.16 \text{ keV}. \quad (1.64)$$

Finally, the three branching ratios are compared in Fig. 1. The denominator is the full Standard Model Higgs width, rather than only the sum of the three channels calculated above. The broad mass scan is taken from the LHC Higgs Cross Section Working Group report CERN-2011-002, while the precise values quoted below at $m_h = 125.2$ GeV use the newer Yellow Report 4 table.

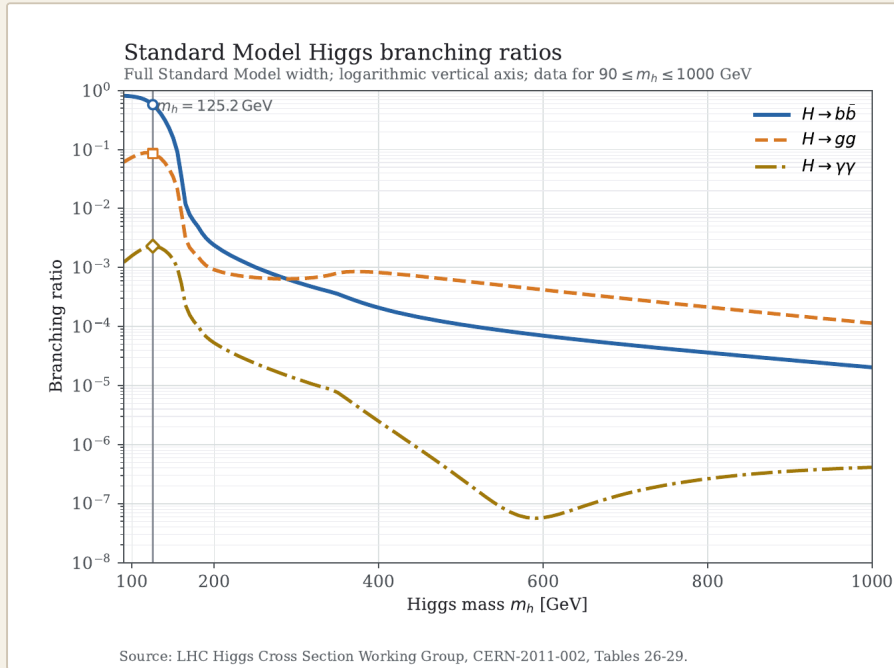


FIGURE 1. Standard Model branching ratios for $H \rightarrow b\bar{b}$, $H \rightarrow gg$, and $H \rightarrow \gamma\gamma$ for $90 \leq m_h \leq 1000$ GeV. The values are taken from Tables 26--29 of CERN-2011-002 and use the full Standard Model decay width. Above approximately 500 GeV, the on-shell Higgs treatment used in these tables should be interpreted with care.

At $m_h = 125.2$ GeV, the corresponding values are

$$\text{BR}(H \rightarrow b\bar{b}) = 0.5792, \quad \text{BR}(H \rightarrow gg) = 0.08172, \quad \text{BR}(H \rightarrow \gamma\gamma) = 0.002270. \quad (1.65)$$